Cleveland Clinic

Boxer’s Fracture

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Boxer’s fractures get their name for how they usually happen: Punching something hard with a closed fist. They’re a type of broken bone, specifically the bone in your hand that connects to your pinkie finger. You’ll probably need to wear a splint or cast and take a break from sports for six to eight weeks.

Overview

Image content: This image is available to view online.

View image online (https://my.clevelandclinic.org/-/scassets/images/org/health/articles/boxers-fracture)

Punching or hitting something hard with a closed fist is the most common cause of boxer’s fractures.

What is a boxer’s fracture?

A boxer’s fracture is a bone fracture (broken bone) in your hand. It’s a type of metacarpal fracture.

Your metacarpals are the bones in your hand that connect your thumb and finger bones (your phalanges) to your wrist. You can feel your metacarpals by pressing on the back of your hand.

A boxer’s fracture is a fifth metacarpal fracture (the bone that connects your pinkie finger to your wrist). It happens when you break the neck of the metacarpal — the end that joins your finger bones.

You’ll probably only need to wear a splint or cast while your bone heals after a boxer’s fracture. More severe fractures require surgery to repair — especially if you have other injuries.

Open vs. closed boxer’s fractures

A healthcare provider will classify your fracture as either open or closed. If you have an open fracture, your bone breaks through your skin. Open fractures usually take longer to heal and have an increased risk of infections and other complications. Closed fractures are still serious, but your bone doesn’t push through your skin.

Displaced boxer’s fractures

”Displaced” or ”nondisplaced” are more words your provider will use to describe your fracture. A displaced fracture means the pieces of your bone moved so much that a gap formed around the fracture when your bone broke (the bone is crooked and not lined up correctly).

Nondisplaced fractures are still broken bones, but the pieces weren’t moved far enough during the break to be out of alignment. Displaced fractures are much more likely to require surgery to repair.

How common are boxer’s fractures?

Boxer's fractures are the most common metacarpal fracture. They're around one-quarter of all broken metacarpals.

Symptoms and Causes

What are boxer's fracture symptoms?

Boxer's fracture symptoms include:

- Pain.
- Swelling.
- Your pinkie finger crossing over or behind your ring finger (a rotational deformity).
- Tenderness.
- Difficulty moving or using your hand.
- Bruising or discoloration.

What causes boxer's fractures?

Punching or hitting something hard with a closed fist (like punching a wall in frustration) is the most common cause of boxer's fractures. They get their name from being a common sports injury that boxers experience.

Getting hit on the back of your hand can also cause a boxer's fracture.

What are the risk factors for boxer's fractures?

Anyone can break their fifth metacarpal bone. Some groups of people are more likely to experience boxer's fractures, including:

- People ages 10 to 40.
- Athletes who play physical sports — especially combat sports like boxing or martial arts.
- People who have osteoporosis.

Diagnosis and Tests

How are boxer's fractures diagnosed?

A healthcare provider will diagnose a boxer's fracture with a physical exam and X-rays.

They'll probably be able to feel or see a boxer's fracture in your hand. Tell your provider what you were doing before you hurt your hand or when you first noticed symptoms.

What tests will be done to diagnose this condition?

Your provider will use X-rays to take pictures of your bones to confirm where your fifth metacarpal is broken and what the fracture looks like.

They may also use a computed tomography (CT) scan to take 3D pictures of your bones and the surrounding tissue (especially if you need surgery).

Management and Treatment

How are boxer's fractures treated?

Your provider will suggest treatments to help your bone heal. Which treatments you'll need depends on which type of boxer's fracture you have and any other injuries you experienced. The most common treatments include:

Immobilization

If the boxer's fracture is mild and your bones didn't move far out of place (if it's nondisplaced), you might only need a splint or cast. Most people need immobilization for three to six weeks. You'll need follow-up X-rays to make sure your bones are healing correctly.

Closed reduction

More severe boxer's fractures may require a closed reduction to set (realign) your bones. Your provider will physically push the outside of your hand to line up your broken bones. To prevent you from feeling pain during the procedure, you'll receive one of the following:

- A local anesthetic to numb the area around your fracture.
- Sedatives to relax your whole body.
- General anesthesia to make you sleep through the procedure.

After the closed reduction, your provider will put your hand in a splint or cast.

Boxer's fracture surgery

Most people don't need surgery for a boxer's fracture.

But your provider may suggest surgery if you have an open fracture, a displaced fracture or a comminuted fracture (if the bone is broken in more than two places). You might also need surgery if the injury damaged tissue other than just your bone. You'll be more likely to need surgery if the fracture causes malrotation (your pinkie finger crosses over your ring finger when you try to bend it).

Your surgeon will realign (set) your bones to their correct position and then, secure them in place so they can heal and grow back together. They usually perform an internal fixation, which means your surgeon will insert metal pins, screws or a plate into your bone to hold it in place. You might need follow-up procedures to remove the fasteners after your bone heals.

Boxer's fracture surgery complications

The most common boxer's fracture surgery complications include:

- **Malunion**: This happens when your broken bones don't line up correctly while they heal.
- **Nonunion**: Your bones may not grow back together fully or at all.
- **Bone infection (osteomyelitis)**: If you have an open fracture (the bone breaks through your skin), you have an increased risk of bacterial infection.
- **Stiffness**: People who experience a boxer's fracture usually feel stiffness around their fractured bone. Home exercises and occupational therapy can help reduce your stiffness.
- **Acute compartment syndrome (ACS)**: A build-up of pressure in your muscles may stop blood from getting to tissue, which can cause permanent muscle and nerve damage.

How soon after treatment will I feel better?

It might take a few weeks for your symptoms to improve. Pain should start getting better in a few days, but it’ll take around six weeks for your bone to regain its full strength.

Depending on which type of immobilization or surgery you need to repair a boxer’s fracture, you should be able to start moving your hand again in a few weeks.

Contact your healthcare provider right away if you experience intense pain that doesn’t get better.

Outlook / Prognosis

What can I expect if I have a boxer’s fracture?

You should make a full recovery if you have a boxer’s fracture.

You’ll need occupational therapy to regain strength and range of motion in your hand.

How long does it take a boxer’s fracture to heal?

Most people need a month or two to recover from a boxer’s fracture. How long it takes you to heal depends on the severity of the fracture and which treatments you need.

There are lots of factors that can affect how long it takes your body to heal. Talk to your healthcare provider or surgeon about a timeline that fits your specific situation.

Will I need to miss work or school with a boxer’s fracture?

You might have to miss work or school while your hand heals, especially if the boxer’s fracture is in your dominant hand (the hand you use most often to write or do other tasks).

Most people can resume physical activities (like playing sports) after eight weeks. Talk to your surgeon or healthcare provider before resuming any physical activities, especially if you broke your bone doing a specific activity or sport.

Prevention

How can I reduce my risk of a boxer’s fracture?

Follow these general safety tips to reduce your risk of an injury:

- Never punch or hit walls, the floor or other hard surfaces when you’re mad or frustrated.
- Always wear your seatbelt.
- Wear the right protective equipment for all activities and sports.
- Follow a diet and exercise plan that helps you maintain good bone health.
- Talk to your provider about a bone density test if you’re older than 50 or if you have a family history of osteoporosis.

Living With

When should I go to the emergency room?

You need to see a healthcare provider as soon as possible if you think you have a boxer’s fracture (or any other type of broken bone). Go to the emergency room if you experience any of the following symptoms:

- Intense pain.
- You can't move your hand or fingers like you usually can.
- Your hand looks noticeably different.
- You can see your bone through your skin.
- Swelling.
- New bruising that appears at the same time as any of these other symptoms.

What questions should I ask my healthcare provider?

- Which type of fracture do I have?
- Will I need surgery?
- How long will I need to wear a splint or cast?
- When is it safe to resume physical activities like playing sports?

Additional Common Questions

Can a boxer's fracture heal on its own?

A boxer's fracture won't heal on its own. A healthcare provider needs to diagnose and treat all bone fractures. If you have an untreated boxer's fracture, you have a much higher risk of complications like malunion.

Malunion happens when a broken bone heals back together, but not in the correct position or alignment. It can make it hard to use your pinkie finger in the future and increases the chances you break that bone again in the future.

Go to the emergency room as soon as you injure your hand and notice any boxer's fracture symptoms. Even if your bone isn't broken, a healthcare provider needs to examine and diagnose your injury.

A note from Cleveland Clinic

A boxer's fracture is a broken bone in your hand. Boxer's fractures get their name for how they usually happen — punching something hard with a closed fist. They're a common sports injury, but can also happen when you punch a wall or other hard surface when you're angry or frustrated.

It's annoying to miss several weeks of practices, meets or other activities, but you should make a full recovery after a boxer's fracture. Don't ignore symptoms like pain and swelling in your hand — especially after an injury. Visit a healthcare provider right away to get a diagnosis. The sooner they diagnose a boxer's fracture, the quicker your broken metacarpal bone can start healing.

Experts You Can Trust

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References

Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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